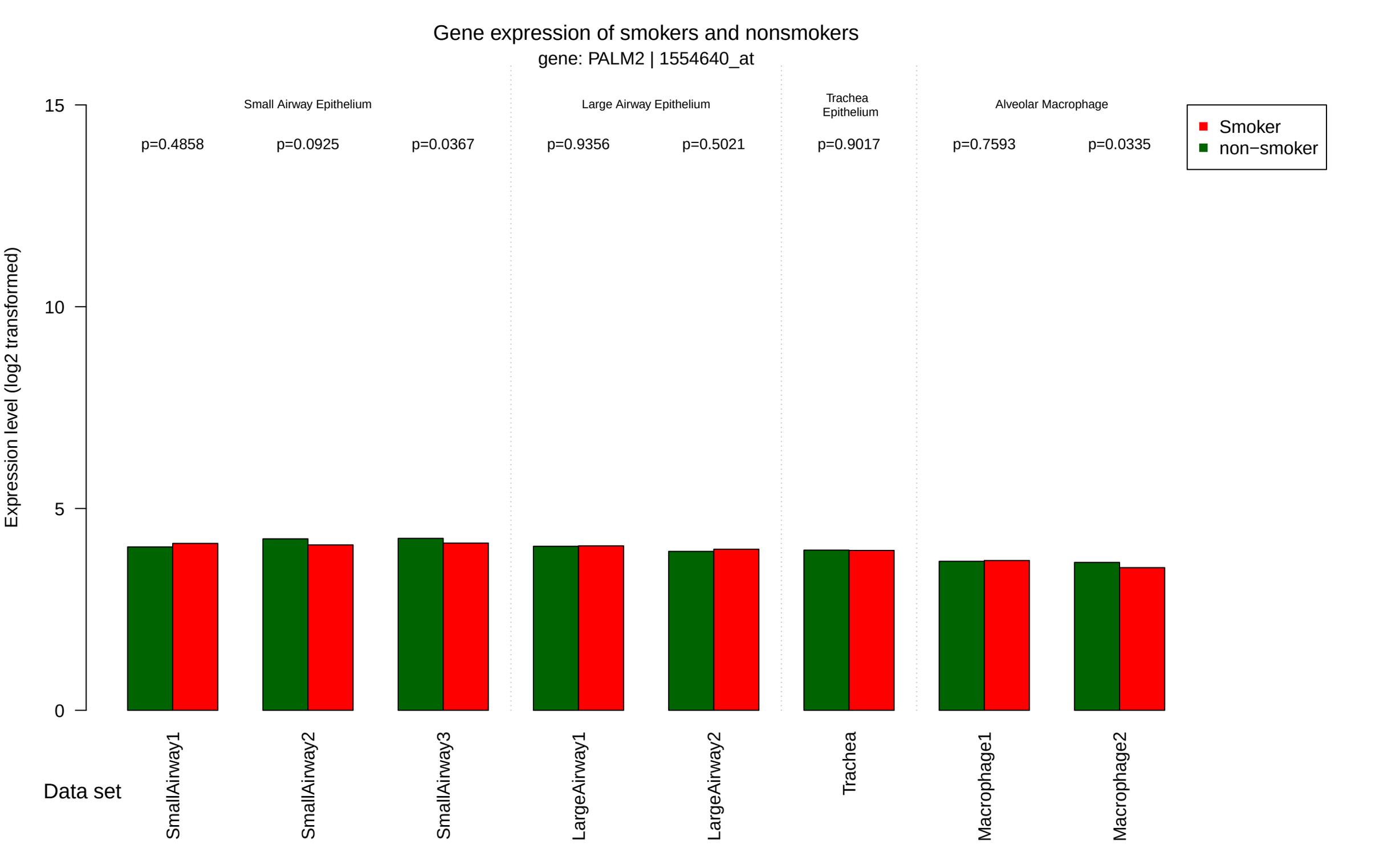
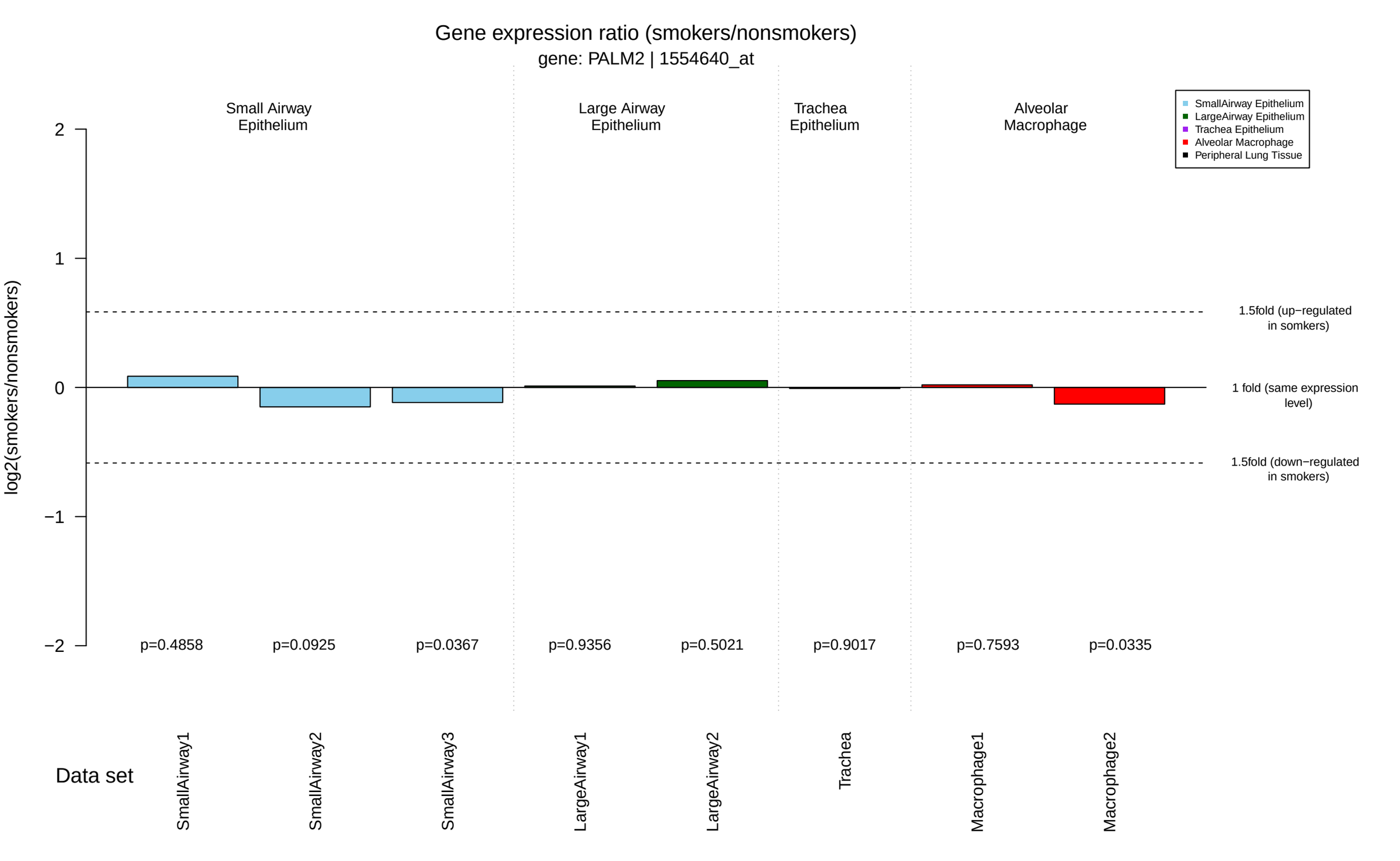
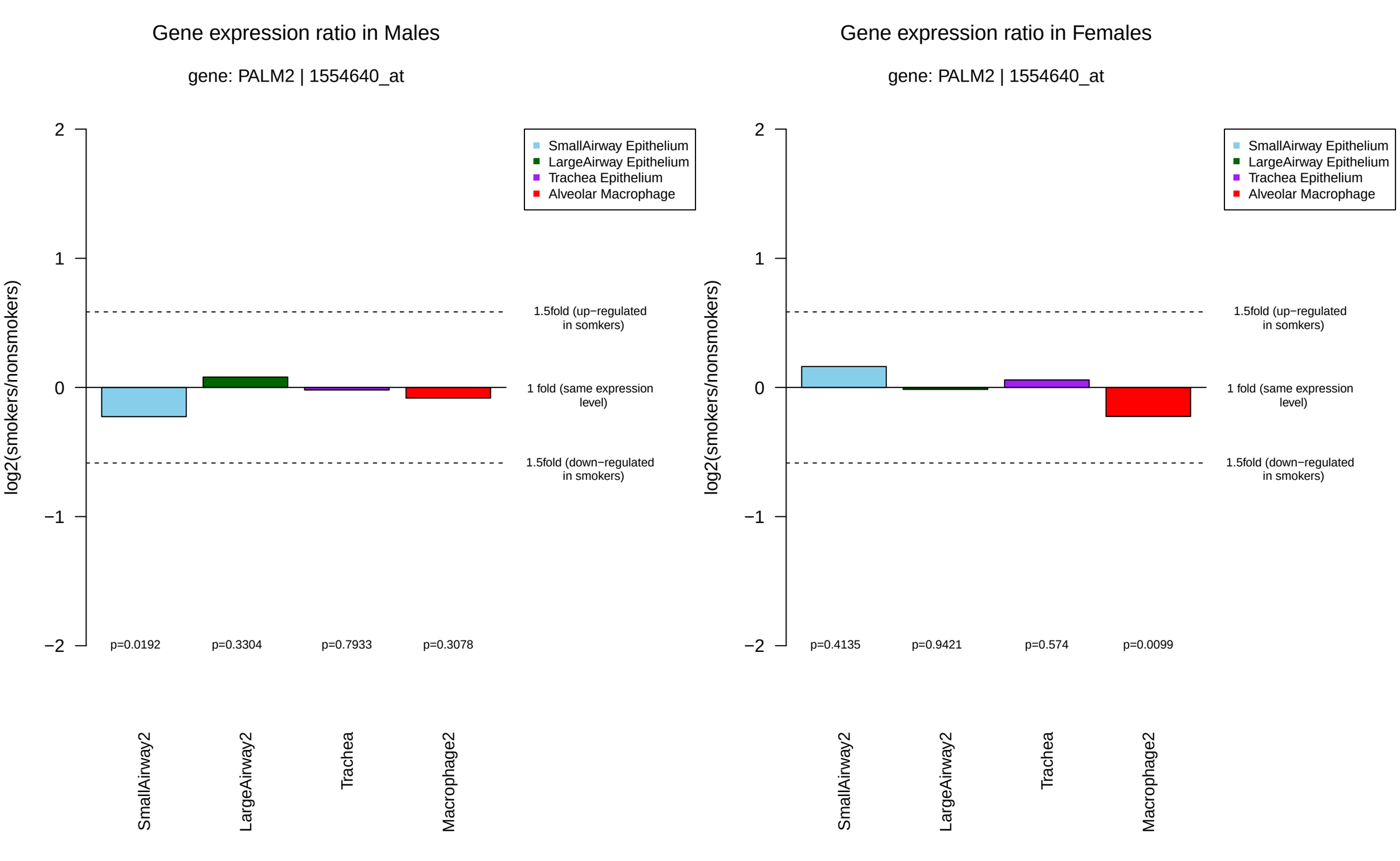
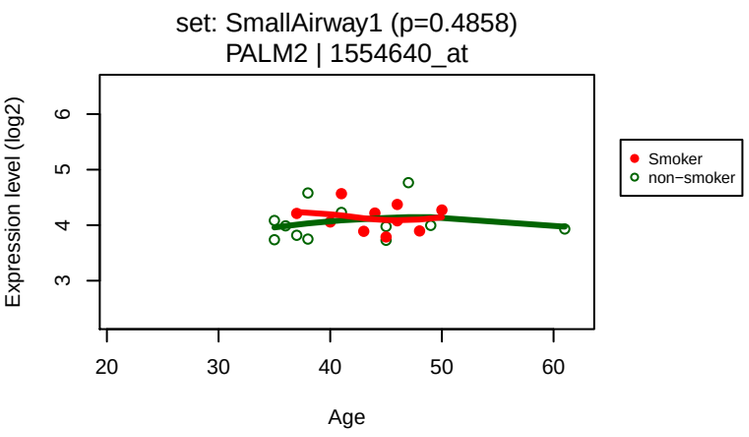
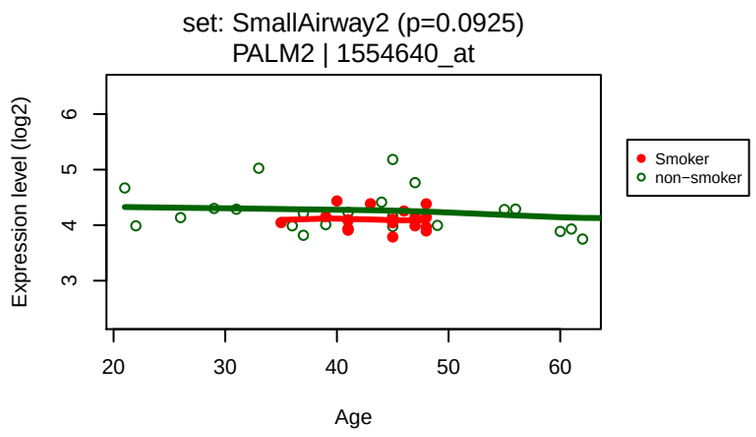
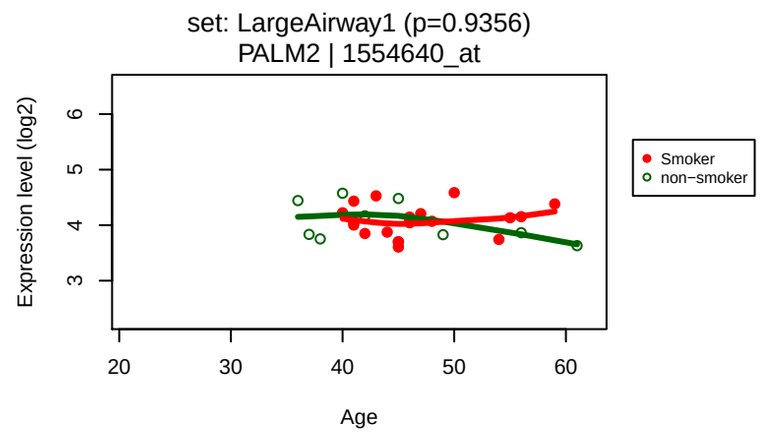

Gene expression of smokers and nonsmokers
gene: PALM2 | 1554640_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4858
p=0.0925
p=0.0367
p=0.9356
p=0.5021
p=0.9017
p=0.7593
p=0.0335
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

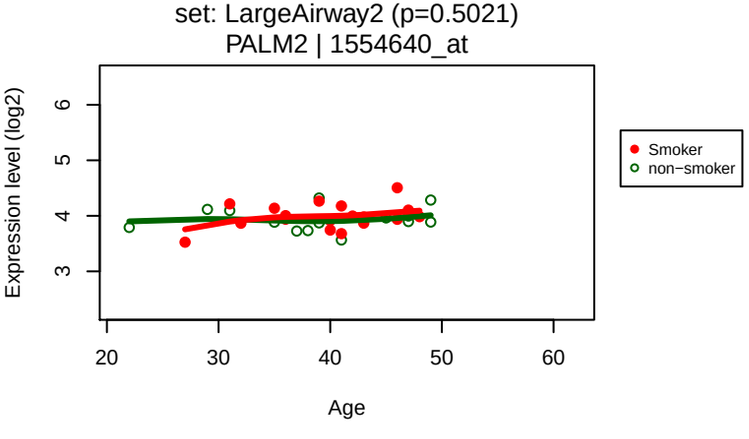

Gene expression ratio (smokers/nonsmokers)
gene: PALM2 | 1554640_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4858
p=0.0925
p=0.0367
p=0.9356
p=0.5021
p=0.9017
p=0.7593
p=0.0335
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

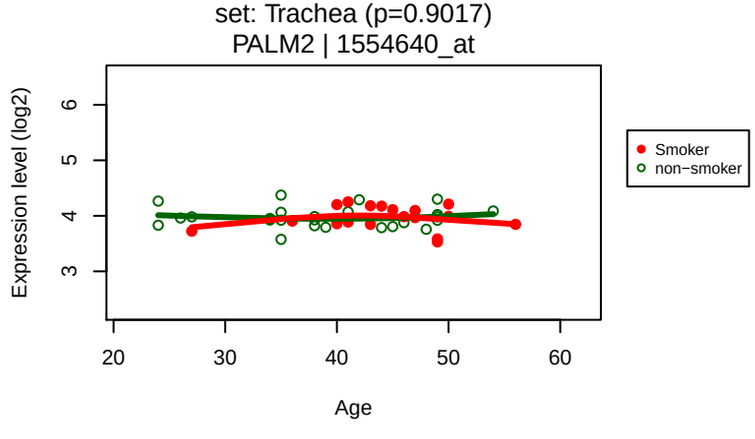

Gene expression ratio in Males
gene: PALM2 | 1554640_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0192
p=0.3304
p=0.7933
p=0.3078
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PALM2 | 1554640_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4135
p=0.9421
p=0.574
p=0.0099
SmallAirway2
LargeAirway2
Trachea
Macrophage2

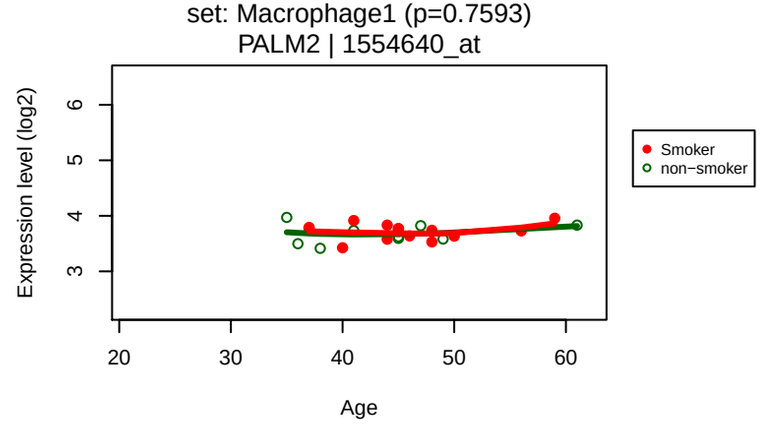

set: SmallAirway1 (p=0.4858)
PALM2 | 1554640_at
Expression level (log2)
Age
Smoker
non−smoker

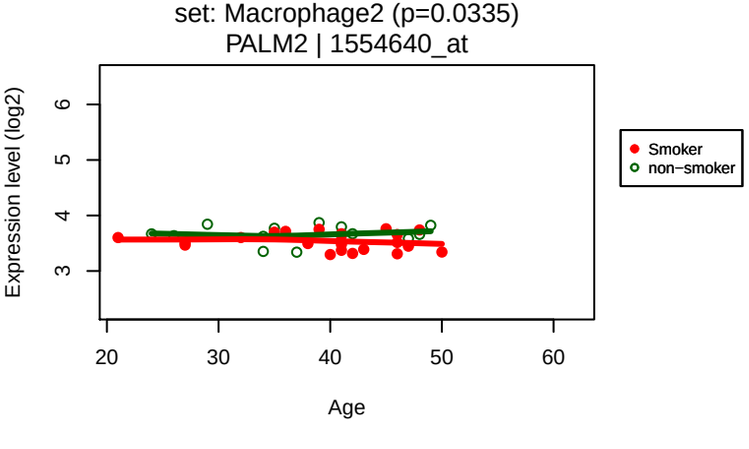

set: SmallAirway2 (p=0.0925)
PALM2 | 1554640_at
Expression level (log2)
Age
Smoker
non−smoker

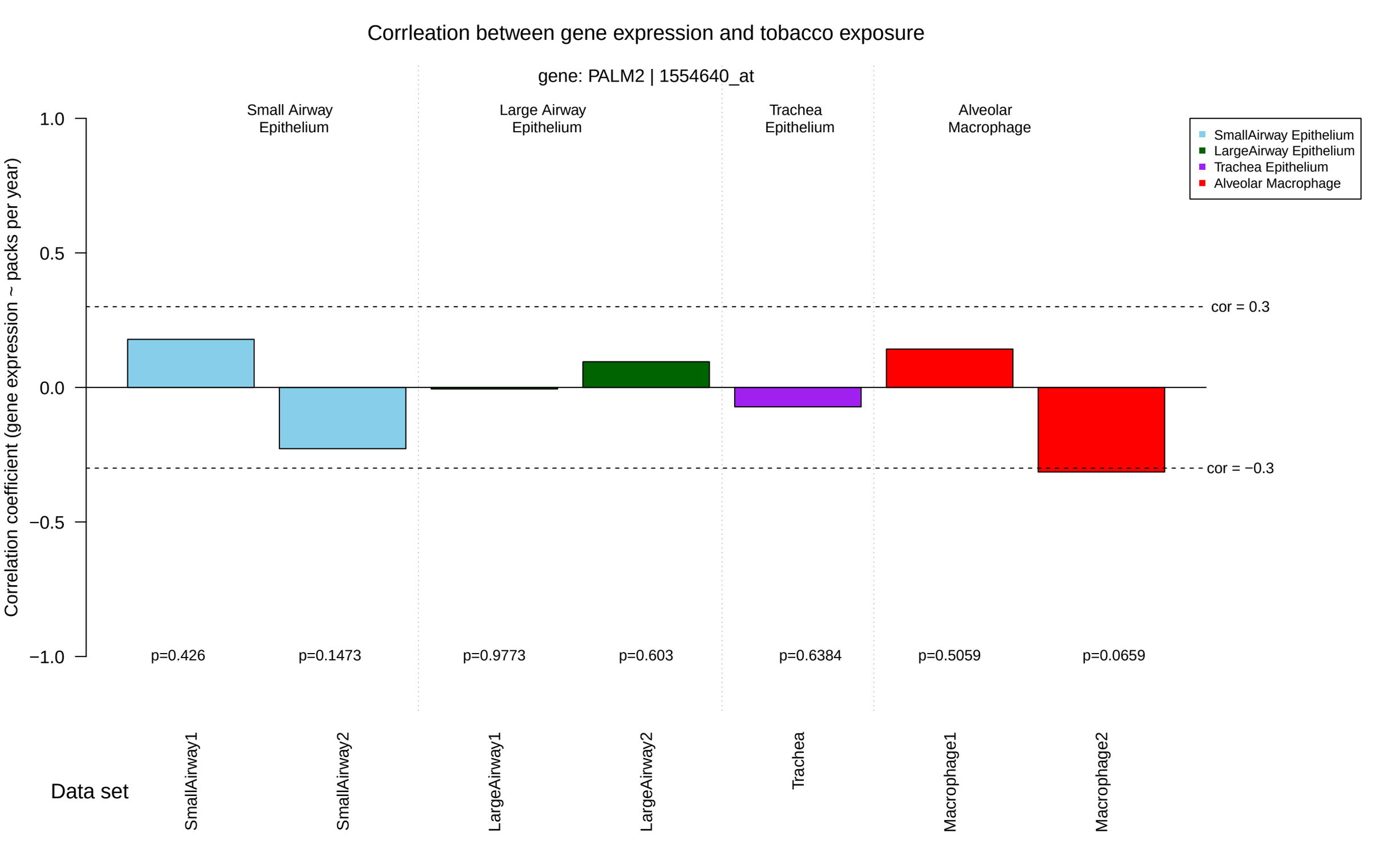

set: LargeAirway1 (p=0.9356)
PALM2 | 1554640_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5021)
PALM2 | 1554640_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.9017)
PALM2 | 1554640_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7593)
PALM2 | 1554640_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0335)
PALM2 | 1554640_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PALM2 | 1554640_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.426
p=0.1473
p=0.9773
p=0.603
p=0.6384
p=0.5059
p=0.0659
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2